W1_report.

Project: Security Posture Assessment and Risk Plan for a Small Organization.

Week 1: 2 to 6 November, 2026.

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GitHub folder: lolacsAcademy/dlh-cyber_security/capstone_security_assessment

1-Summary:

Week 1 covered the scope and the asset inventory, as planned. The lab is ready: the 3 machines are installed and their audit tools are in place. The organization profile and the asset inventory are complete, and week 2 can start with the audits.

2-Work done

2.1-Lab environment

Preparation started before the official start date, so that week 1 could focus on the scope and the inventory.

ID	Machine	System	Audit tool
M01	Windows 11 VM (employee laptop sample)	Windows 11 Enterprise Evaluation, build 26300.	HardeningKitty 0.9.4
M02	Ubuntu VM (file server)	Ubuntu 24.04.5 LTS	Lynis 3.0.9
M03	Kali VM (audit workstation)	Kali GNU/Linux Rolling	Lynis 3.1.6

• Each machine has a saved snapshot (a restore point), so it can be returned to a working state if anything goes wrong.

• Both tools work in read-only mode: no settings are changed on any machine.

• HardeningKitty was test-run once on M01 to confirm it works. Those test results are not used; the real audit runs in week 2.

Figure 1- VirtualBox list of the three project machines.

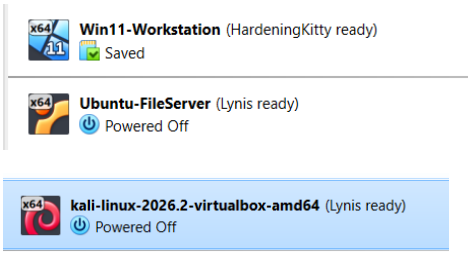


Figure 2- Lynis version on Kali 3.1.6

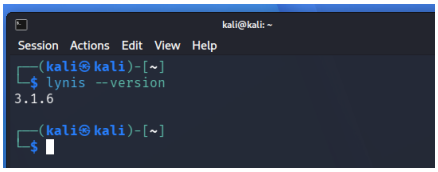


Figure 3- Lynis version on Ubuntu 3.0.9

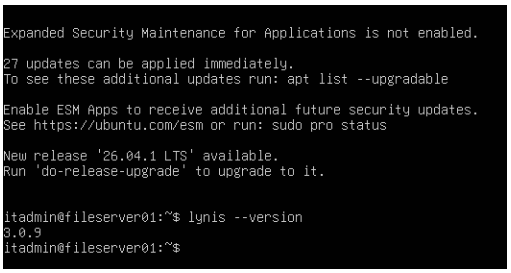
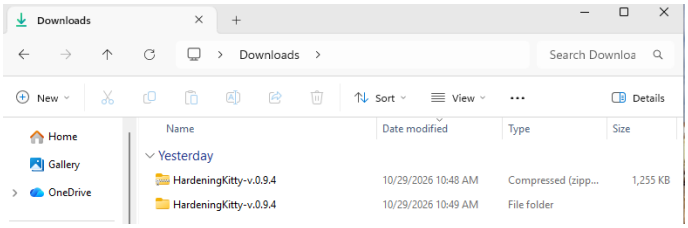


Figure 4- HardeningKitty folder on Windows (v.0.9.4)



2.2 Organization profile and scope

Full document: W1_Organization_Profile_and_Scope (on GitHub)

Summary

• Organization: Alder & Co Accounting (fictional), Luxembourg, 10 employees.

• IT environment: 10 Windows 11 laptops with one standard setup, one Ubuntu file server, email and accounting software run by external providers.

• Scope: three machines audited (Windows 11, Ubuntu, Kali), read-only, against NIST CSF 2.0 and CIS Controls v8.

• Out of scope: email and accounting software (external providers), router, Wi-Fi and printer.

The firm remains responsible for the data these providers hold.

2.3 Asset Inventory

Full document: W1_Asset_Inventory (on GitHub).

Summary

Each asset is rated High, Medium or Low for Confidentiality, Integrity and Availability. Its criticality is the highest of the three ratings. The facts about each machine were collected directly from the machines with read-only commands.

ID	Asset	Criticality
M01	Employee laptops (sample: DESKTOP-TVVAHP9)	High
M02	File server (fileserver01)	High
M03	Kali audit workstation	Not rated (not an asset of the organization)
D01	Payroll data and staff personal details	High
D02	Client tax and financial records	High
D03	Internal procedures and templates	Medium
D04	Website content and brochures	Medium
S01-S03	Email, accounting software, office network	High (outside the audit scope)

Main result: the file server (M02) holds both payroll and client records, so its audit findings will be reviewed first in week 2.

Figure 5- Windows facts (Get-CimInstance)

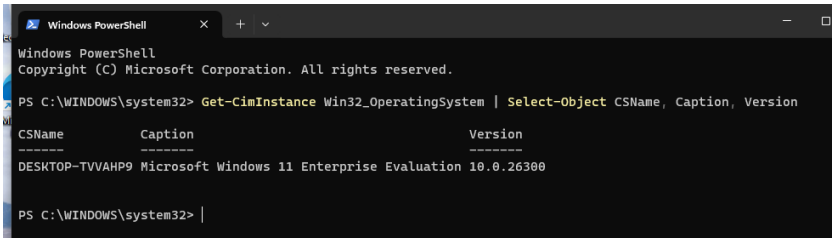


Figure 6- Ubuntu facts (hostnamectl)

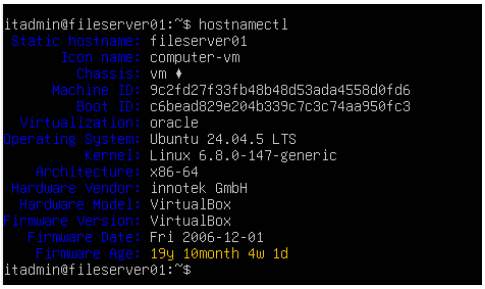
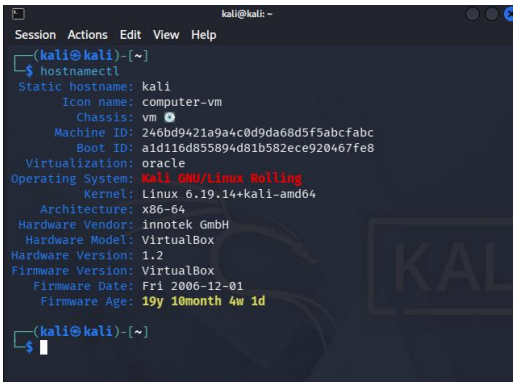


Figure 7- Kali facts (hostnamectl)



3- Issues met and they were solved

• Slow virtual machines: VirtualBox was running in a slow mode because two Windows features on the host laptop (Memory integrity and Virtual Machine Platform) were using the same processor function. Both were turned off for the project period, and VirtualBox returned to normal speed (its status changed from “native API” to “VT- x/AMD-V”. Memory integrity will be turned back on after the project.

• CIS list version: the Windows VM runs build 26300, which is newer than the latest CIS Windows 11 list included in HardeningKitty 0.9.4 (24H2). The 24H2 list is used as the closest available, and this is stated in the audit results.

• Lynis version: each Linux system installs its own Lynis version (3.0.9 on Ubuntu, 3.1.6 on Kali). The version used is recorded for each machine.

4-Feedback from the pitch and how it was applied

Feedback (recap email, 27 October)	Applied in week 1
Audit not limited to Kali	Windows 11 and Ubuntu VMs added; kali also kept in the audit scope
Ten computers in the organization	One representative Windows 11 laptop stands for the ten (same standard setup)
Tangible results	Real facts collected from each machine, with screenshots
Storage on GitHub	Week 1 files uploaded to GitHub, folder capstone_security_assessment

5- plan for week 2 (9 to 13 November)

• Run HardeningKitty on M01, and Lynis on M02 and M03.

• Save the result files and screenshots of each audit.

• Start the gap analysis against CIS Controls v8, beginning with M02.